

# A binomial sharpening of the cube–BMO rearrangement bound

Autonomous proof packet for arXiv:2011.09111

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**Status.** Partial result, likely valid. The best constants and the dimension-free question remain open.

## 1 Source question

Burchard, Dafni, and Gibara prove that every rearrangeable  $f \in \text{BMO}(\mathbb{R}^n)$  satisfies

$$\|f^*\|_{\text{BMO}(\mathbb{R}_+)} \leq 2(1 + 2\sqrt{n-1})\|f\|_{\text{BMO}(\mathbb{R}^n)}. \quad (1)$$

On page 3 they ask for the best constants  $C_n$  and whether the constants can be dimension-free for  $n > 1$  [1].

While the sharp constant  $C_1 = 1$  of Klemes–Korenovskii holds in dimension  $n = 1$ , the question of the best constants  $C_n$  in Eq. (1.1), and whether such constants can be dimension-free, remains open for  $n > 1$ . However, sharp results are known for other BMO-spaces. For the smaller space  $\text{BMO}_{\mathcal{R}}$ , called anisotropic or strong BMO, consisting of functions of bounded mean oscillation on rectangles, a sharp constant of 1 was obtained by Korenovskii in [20] by using a higher-dimensional version of the Riesz Rising Sun lemma for rectangles - see also Korenovskii–Lerner–Stokolos [22].

For a finite-measure set  $S$ , write

$$f_S = \frac{1}{|S|} \int_S f, \quad \mathcal{O}(f, S) = \frac{1}{|S|} \int_S |f - f_S|.$$

The usual BMO seminorm is the supremum of this quantity over axis-parallel cubes.

## 2 Idea of the proof

The  $\sqrt{n}$  term in (1) enters at one precise point. A false cube with  $m$  long directions is tiled by  $2^m$  ordinary cubes. Their means form a function  $g$  on the Bernoulli cube  $\{0, 1\}^m$ , and neighboring values of  $g$  differ by at most  $4\|f\|_{\text{BMO}}$ . The source applies a variance bound and Cauchy–Schwarz to obtain  $\mathbb{E}|g - \mathbb{E}g| \leq 2\sqrt{m}\|f\|_{\text{BMO}}$ .

The key observation is that the Doob differences of  $g$  are symmetric two-point variables with predictable amplitudes. For an even convex loss, backward conditioning can only increase its expectation when every amplitude is replaced by its maximum. Thus the exact comparison variable

is a Rademacher sum, not a Gaussian second-moment upper bound. Its  $L^1$  norm is the mean absolute deviation of a symmetric binomial variable and is smaller by the asymptotic factor  $\sqrt{2/\pi}$ .

A separate geometric lemma says that the oscillation on a subset occupying fraction  $\alpha$  of a set is at most  $\alpha^{-1}$  times the oscillation on the containing set. Circumscribing a false cube supplies a second bound; taking the better bound for each number of long directions gives the stated hybrid constant.

### 3 Sharp Bernoulli concentration

**Lemma 1** (convex domination). *Let  $X_1, \dots, X_m$  be independent unbiased Bernoulli variables, and let  $g = g(X_1, \dots, X_m)$ . Suppose changing coordinate  $i$  while holding all others fixed changes  $g$  by at most  $a_i$ . If  $\varepsilon_1, \dots, \varepsilon_m$  are independent uniform signs, then for every even convex  $\Phi : \mathbb{R} \rightarrow \mathbb{R}$  for which the expectations exist,*

$$\mathbb{E}\Phi(g - \mathbb{E}g) \leq \mathbb{E}\Phi\left(\frac{1}{2} \sum_{i=1}^m a_i \varepsilon_i\right).$$

In particular,

$$\mathbb{E}|g - \mathbb{E}g| \leq \mathbb{E}\left|\frac{1}{2} \sum_{i=1}^m a_i \varepsilon_i\right|. \quad (2)$$

The bound is sharp when  $g = \sum_i a_i X_i$ .

*Proof.* Let  $M_i = \mathbb{E}[g \mid X_1, \dots, X_i]$  be the Doob martingale. Conditional on  $X_1, \dots, X_{i-1}$ , the two possible values of  $M_i$  have difference  $c_i = c_i(X_1, \dots, X_{i-1})$  with  $|c_i| \leq a_i$ : averaging over the future coordinates preserves the bounded-difference hypothesis. Hence, after identifying  $2X_i - 1$  with  $\varepsilon_i$ ,

$$g - \mathbb{E}g = \frac{1}{2} \sum_{i=1}^m c_i(\varepsilon_1, \dots, \varepsilon_{i-1}) \varepsilon_i.$$

For fixed  $s$ , convexity implies that

$$t \mapsto \frac{\Phi(s+t) + \Phi(s-t)}{2}$$

is nondecreasing for  $t \geq 0$ . Condition first on  $\varepsilon_1, \dots, \varepsilon_{m-1}$  and replace  $|c_m|$  by  $a_m$ . After averaging in  $\varepsilon_m$ , the resulting function of the partial sum is again even and convex. Repeat backward for  $m-1, m-2, \dots, 1$ . This gives the claimed domination. Taking  $\Phi(t) = |t|$  proves (2); constant amplitudes give equality.  $\square$

Let  $B_m \sim \text{Bin}(m, \frac{1}{2})$  and put

$$\beta_m := \mathbb{E}|B_m - m/2| = \frac{m}{2^m} \binom{m-1}{\lfloor (m-1)/2 \rfloor}. \quad (3)$$

The identity follows by splitting the binomial sum at its mean and using  $k \binom{m}{k} = m \binom{m-1}{k-1}$ . It also shows  $\beta_{2r-1} = \beta_{2r}$  and that  $(\beta_m)$  is nondecreasing. Stirling's formula gives

$$\beta_m = \sqrt{\frac{m}{2\pi}} + O(m^{-1/2}). \quad (4)$$

## 4 Application to false cubes

A false cube in  $\mathbb{R}^n$  has side lengths  $2s$  in  $m$  coordinate directions and  $s$  in the other  $n - m$  directions. Denote the family of all false cubes by  $\mathcal{W}$ .

**Theorem 1** (binomial false-cube comparison). *If  $R$  is a false cube with  $m < n$  long directions, then*

$$\mathcal{O}(f, R) \leq (1 + 4\beta_m)\|f\|_{\text{BMO}(\mathbb{R}^n)}. \quad (5)$$

Consequently,

$$\|f\|_{\text{BMO}_{\mathcal{W}}} \leq (1 + 4\beta_{n-1})\|f\|_{\text{BMO}}. \quad (6)$$

The term  $4\beta_m$  in (5) is optimal as a consequence of only the individual neighboring-cube bounds employed below.

*Proof.* Tile  $R$  by its  $2^m$  congruent cubes  $Q(\nu)$ , indexed by  $\nu \in \{0, 1\}^m$ , and set  $g(\nu) = f_{Q(\nu)}$ . The partition estimate in [1, Lemma 3.6] gives

$$\mathcal{O}(f, R) \leq \|f\|_{\text{BMO}} + \mathbb{E}|g - \mathbb{E}g|.$$

Means on adjacent equal cubes differ by at most  $4\|f\|_{\text{BMO}}$  [1, Lemma 3.7]. Lemma 1, with every  $a_i = 4\|f\|_{\text{BMO}}$ , yields

$$\mathbb{E}|g - \mathbb{E}g| \leq 2\|f\|_{\text{BMO}} \mathbb{E} \left| \sum_{i=1}^m \varepsilon_i \right| = 4\beta_m \|f\|_{\text{BMO}}.$$

This proves (5); monotonicity of  $\beta_m$  gives (6). The additive Hamming-weight function attains Lemma 1, proving the final assertion about this information set.  $\square$

Combining Theorem 1 with the dimension-free factor 2 for the decreasing rearrangement on the bisection basis  $\mathcal{W}$  [1, Lemma 3.5] immediately gives

**Corollary 1.** *For  $n \geq 2$  and every rearrangeable  $f \in \text{BMO}(\mathbb{R}^n)$ ,*

$$\|f^*\|_{\text{BMO}(\mathbb{R}_+)} \leq 2(1 + 4\beta_{n-1})\|f\|_{\text{BMO}(\mathbb{R}^n)}. \quad (7)$$

*In particular the leading coefficient in the source bound is improved by  $\sqrt{2/\pi}$ :*

$$C_n \leq \sqrt{\frac{32(n-1)}{\pi}} + 2 + O(n^{-1/2}).$$

## 5 A geometric hybrid refinement

**Lemma 2** (subset-to-container oscillation). *Let  $E \subset Q$  have  $0 < |E| < |Q|$ , and put  $\alpha = |E|/|Q|$ . Then*

$$\mathcal{O}(f, E) \leq \alpha^{-1} \mathcal{O}(f, Q).$$

*Proof.* Write  $u = f_E$ ,  $v = f_{Q \setminus E}$ , and  $w = f_Q = \alpha u + (1 - \alpha)v$ . Jensen's inequality on  $Q \setminus E$  gives

$$\begin{aligned} \mathcal{O}(f, Q) &\geq \alpha \int_E |f - w| + (1 - \alpha) \int_{Q \setminus E} |f - w| \\ &= \alpha \left( \int_E |f - w| + \int_{Q \setminus E} |u - w| \right) \\ &\geq \alpha \int_E |f - u|. \end{aligned}$$

The last step is the triangle inequality.  $\square$

A false cube with  $m$  long directions lies in a cube of side  $2s$  and occupies the fraction  $2^{m-n}$  of that cube. Lemma 2 therefore supplies the second estimate

$$\mathcal{O}(f, R) \leq 2^{n-m} \|f\|_{\text{BMO}}. \quad (8)$$

**Theorem 2** (hybrid rearrangement bound). *For  $n \geq 2$ , define*

$$K_n = \max_{1 \leq m \leq n-1} \min\{1 + 4\beta_m, 2^{n-m}\}.$$

*Then every rearrangeable  $f \in \text{BMO}(\mathbb{R}^n)$  satisfies*

$$\|f^*\|_{\text{BMO}(\mathbb{R}_+)} \leq 2K_n \|f\|_{\text{BMO}(\mathbb{R}^n)}. \quad (9)$$

*Proof.* For each false cube choose the better of (5) and (8), then maximize over its number  $m$  of long directions. Ordinary cubes (the case  $m = n$ ) cost only 1 and do not enlarge  $K_n$ . Apply the factor-2 bisection-basis rearrangement theorem from [1].  $\square$

For illustration,  $2K_n$  for dimensions 2 through 12 is

$$4, 6, 6, 8, 8, 8, 9.5, 9.5, 10.75, 10.75, 11.84375.$$

The clean bound (7) already improves the asymptotic leading constant. The hybrid estimate is never larger, is often strictly smaller, and also uses only elementary geometry.

## 6 Verification and limitations

The accompanying script verifies formula (3), monotonicity through  $m = 200$ , the displayed  $K_n$  values, and the comparison with the source constant. It also solves the finite linear programs defining the maximum mean absolute deviation of a 1-Lipschitz function on  $\{0, 1\}^m$  for  $m \leq 3$ ; the optima equal  $\beta_m$ . These computations are checks only; the proof above is analytic.

The result does not decide whether  $(C_n)$  is dimension-free and does not determine any sharp  $C_n$  for  $n > 1$ . Moreover, sharpness of the discrete lemma shows that a dimension-free upgrade cannot use only separate bounds on the edges of the cube of subcube means. It would need additional geometric compatibility among those means or a different decomposition.

A bounded novelty search on 11 August 2026 covered the run indexes, exact title and inequality queries, author/citation searches, and combinations of “false cube”, “binomial”, “mean absolute deviation”, and “dimension-free”. The later VMO work by the same authors and related semigroup-BMO results do not sharpen this cube-BMO constant. No occurrence of the present application was found. Novelty confidence is moderate because Lemma 1 is elementary and may be known in martingale language.

**Human-review recommendation.** Check the backward induction in Lemma 1, the factor  $4\beta_m$  in Theorem 1, and the normalization in Lemma 2. Subject to those checks, classify this as a substantial partial improvement, not a full resolution.

## References

- [1] A. Burchard, G. Dafni, and R. Gibara, *Mean oscillation bounds on rearrangements*, Trans. Amer. Math. Soc. 375 (2022), 4429–4444; arXiv:2011.09111.